

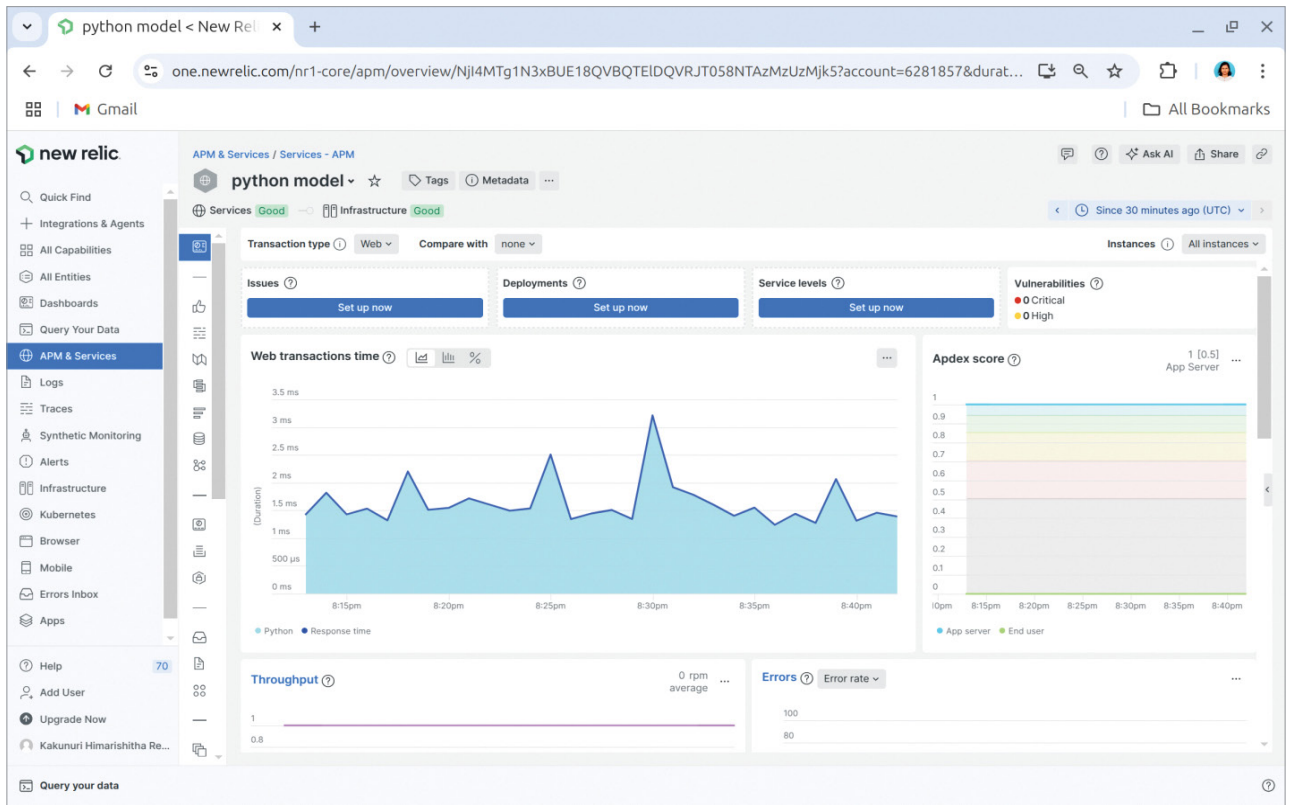


Figure 3: New Relic dashboard displaying key Python application performance metrics, including response times, throughput, and error rates

Browser monitoring for the frontend application

To ensure an optimal user experience, New Relic browser monitoring was integrated into the frontend application. This monitoring solution helps track real-user interactions, page load times, JavaScript errors, and frontend performance metrics, providing deep insights into how the application behaves in the user's browser.

Since the frontend application is built using React and deployed in a Docker container, the New Relic browser agent was added to capture real-time performance analytics. The following steps outline the integration process.

Step 1: Enable browser monitoring in New Relic

- Navigate to *Integrations & Agents* in the New Relic dashboard.
- Select *Browser Monitoring* as the integration type.

Step 2: Choose an integration method

- Choose how the New Relic browser agent should be integrated.
- Click *Continue* to proceed.

Step 3: Configure the browser monitoring agent

- Assign a name to the frontend application for tracking in New Relic.

- Define the monitoring settings based on the application's needs.

Step 4: Add the New Relic agent script to the HTML file

- Copy the JavaScript snippet provided by New Relic.
- Insert the script into the `<head>` section of the HTML template in the frontend application.

Step 5: Build and deploy the frontend application

- Build the Docker image for the frontend:

```
docker build -t my-frontend-app
```

- Run the Docker container:

```
docker run -d -p 3000:3000 my-frontend-app
```

- Navigate to the New Relic browser monitoring section and verify that the frontend application appears in the dashboard as shown in Figures 4 and 5.

Key performance metrics captured

Once the browser monitoring integration is complete, New Relic collects real-time data on the following:

- *Page load times:* Measures how quickly pages render for users.